

Real-Time DIGITAL CLOCK

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A real-time digital clock (RTC) is often required in projects and designs where we need to deal with time. RTC helps in keeping track of time even when the main processor is asleep or switched off. It is because of RTC that your PC/laptop clock keeps showing

the correct time even after it has been switched off for some time, or even when the computer's battery is removed for replacement with a new one. RTC module has its own small coin cell that runs for years and helps keep track of time when the whole circuit is powered off.

Let us try to design our own real-time digital clock that can be used in time-related projects, like keeping the data of a sensor with time, or a time based alarm system, or a power cut-off system. The components required for the circuit are listed in the Bill of Material table.

This project aims at building a digital clock using a real-time clock chip (DS3231) with battery backup, a liquid crystal display (JHD 204A), and Arduino Uno board. You will also learn how to program the DS3231 clock chip and the LCD in an Arduino environment.

The project has two parts. In the first part, we learn how to interface the LCD to Arduino Uno. In the second part, we add a clock chip to this setup and display the time.

This RTC clock is to be made using Arduino Uno and programmed in Arduino IDE. So, download Arduino IDE and unzip it in a convenient folder. Fig. 1 shows the Arduino Uno board.

Fig. 2(a) shows the display side (front side) of the LCD, while Fig. 2(b) shows its back side. The time is dis-

played on liquid crystal display JHD 204A. This is a 20-column, 4-row display with blue backlight but you can also use a 16 × 2 LCD with two rows of the display. Each character is displayed in a dot-matrix manner.

The LCD has 16 pins but only 12 pins are used here. Out of these, 4 are for data lines, 3 for control lines, and the rest are for power and ground lines. (We are not using DB0, DB1, DB2, and DB3 here.)

Pin No. 3 is for controlling the contrast. A 10k trim pot is connected to this pin. Adjust this pot to get optimum performance. Pin No. 15 of the LCD is connected to 5V through a 220-ohm resistor.

If you are using the SPI to I2C converter with display, you got the four pins SCL, SDA, GND, and VCC. Here in this design, SPI is used.

BILL OF MATERIAL

Components	Quantity
RTC DS3231 (MOD1)	1
CR2032 coin cell	1
LCD JHD 204A / 1602 I2C / SPI (MOD2)	1
10k pot (P1)	1
220Ω resistor (R1)	1
Arduino Uno with USB cable	1
Power adaptor (12V, 2A)	1
Jumper wires (male-female)	20



Fig. 3: RTC DS3231



Fig. 1: Arduino Uno board



Fig. 2(a): Front side of LCD



Fig. 2(b): Back side of LCD